

Gold Price Prediction

Methodology, Results & Conclusions

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Abstract. This report documents the full methodology of a machine learning project aimed at forecasting gold (XAU/USD) prices using multi-source financial data. The pipeline integrates 25+ years of daily market prices from Yahoo Finance with macroeconomic indicators from the Federal Reserve (FRED), enriched with 40+ engineered technical features. Four models were trained and evaluated: Ridge Regression, Lasso Regression, XGBoost, and LightGBM. Ridge and Lasso substantially outperformed the tree-based models, with both achieving $R^2 = 0.9973$ on Hanaa's experiment and Ridge achieving $R^2 = 0.9851$ on Yosef's. This report explains the reasoning behind each decision — from data source selection to feature engineering choices to model evaluation strategy.

0.9973

Best R^2 (Ridge & Lasso)

26.0

Best MAE — \$USD (Lasso)

46.7

Best RMSE — \$USD (Lasso)

4

Models Evaluated

69

Features

6,419

Trading Days

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1 Introduction & Problem Statement

Gold is one of the world's most strategically significant assets — serving simultaneously as an inflation hedge, safe-haven instrument, and macroeconomic indicator. Despite this importance, gold price forecasting remains a non-trivial problem. Its price is influenced by a wide range of factors: monetary policy decisions, currency fluctuations, geopolitical risk, investor sentiment, and physical demand from countries like India and China.

Traditional approaches — such as simple moving average models or ARIMA — treat gold as an isolated time series and miss the cross-asset relationships that drive its price. This project takes a different approach: we treat gold price prediction as a supervised machine learning problem, feeding the model a rich set of features from multiple sources so it can learn these relationships from the data.

The central question is: **can a machine learning model trained on market, macroeconomic, and technical features accurately predict the next day's gold price?**

2 Data Collection

Source Selection

We collected data from two complementary sources. The first is **Yahoo Finance**, accessed via the `yfinance` Python library, which provides daily closing prices for 19 financial instruments including gold futures (GC=F), silver, crude oil, the US Dollar Index (DXY), equity indices (S&P; 500, Nasdaq), volatility (VIX), Treasury yields, and exchange rates. These represent the market-level forces that move gold on a daily basis.

The second source is **FRED (Federal Reserve Economic Data)**, accessed via `pandas-datareader` without requiring an API key. FRED provides 10 official macroeconomic series including CPI (inflation), the Federal Funds Rate, Real 10-Year Interest Rate, M2 Money Supply, Producer Price Index, Treasury Yield Spread, Unemployment Rate, GDP Growth, Industrial Production, and Consumer Sentiment.

Why Both Sources?

Yahoo Finance captures what is *happening right now* in markets — the instantaneous pricing of risk, sentiment, and liquidity. FRED captures the *underlying economic forces* that drive gold over longer cycles. A model trained on only one source would miss half the picture. For example, the DXY (dollar index) explains short-term gold moves, while real interest rates and M2 money supply explain the multi-year bull and bear cycles.

Data Alignment

FRED data is published monthly or quarterly — not daily. To align it with the Yahoo Finance daily data, we applied forward-filling: each monthly reading is carried forward to fill every trading day in that month until the next official reading is released. This is the standard approach in quantitative finance and avoids look-ahead bias since we only use data that would have been available at the time.

Source	Tickers / Series	Frequency	Features
Yahoo Finance	GC=F, SI=F, CL=F, DXY, VIX, ^GSPC, ^TNX, EURUSD=X ...	Daily	19
FRED	CPI, FEDFUNDS, M2SL, UNRATE, T10Y2Y, REAINTRATREARAT10Y ...	Monthly	10
Engineered	RSI, MACD, Bollinger Bands, MA/EMA, Returns, Lags, Ratios	Daily	40+

Before building any model, we conducted a thorough EDA to understand the statistical properties of the data and identify which features carry the strongest signals.

Key EDA Findings

- **Autocorrelation dominates.** Gold price shows near-perfect autocorrelation in the short term — yesterday's price is the single strongest predictor of today's price. This is reflected in lag features and moving averages showing correlation > 0.998 with the target.
- **Non-normal return distribution.** Daily returns exhibit negative skewness and fat tails (excess kurtosis), meaning extreme events occur more frequently than a normal distribution would predict. This informed our choice of robust evaluation metrics.
- **DXY is the strongest independent signal.** The US Dollar Index shows a consistent -0.68 correlation with gold price — the strongest cross-asset signal that is not derived from gold itself.
- **Real interest rates are the key macro driver.** The Real 10-Year Rate showed -0.51 correlation, consistent with academic finance literature showing that gold performs best when real rates are negative or falling.
- **M2 Money Supply and CPI are highly predictive.** Both showed correlations above 0.95 with gold price, reflecting gold's role as an inflation hedge over multi-year cycles.
- **Seasonality is measurable.** Monthly analysis showed consistent patterns: September–November bullish (Indian festival season), June bearish (summer demand lull).
- **88 extreme outlier days** were identified ($Z\text{-score} > 3$), each corresponding to major macro events: Lehman collapse (2008), COVID-19 (2020), and geopolitical shocks (2025–2026).



Figure 1: Stylized gold price trajectory 2001–2026 with key macro events annotated.

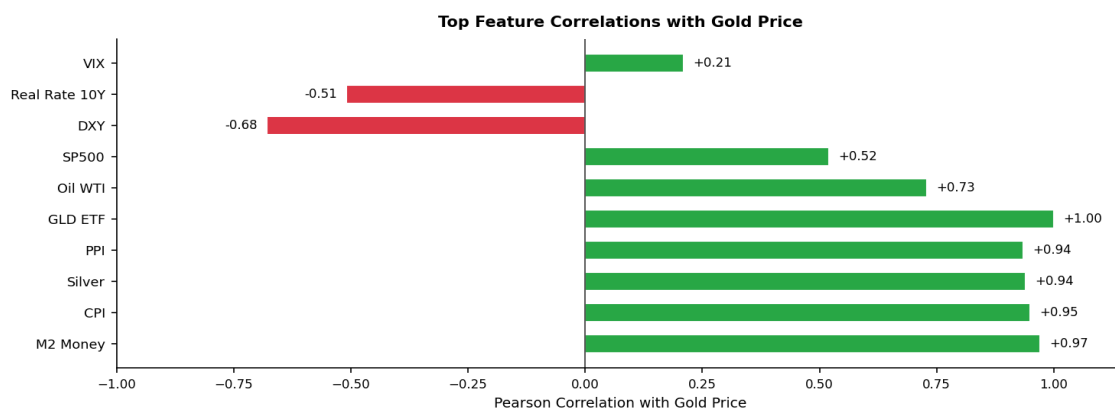


Figure 2: Pearson correlation of top features with gold price. Green = positive, Red = negative.

Missing Value Treatment

Two categories of missing values required different treatments. FRED macro series were forward-filled as described above. Technical indicators with warm-up periods (MA200 requires 200 days, RSI requires 14 days) produced NaN in early rows — these were removed by dropping the first 200 rows of the dataset, ensuring every training sample has a complete feature vector.

Feature Engineering Decisions

We engineered features across five categories, each serving a specific purpose:

Feature Group	Purpose & Reasoning
Moving Averages (MA & EMA)	Capture trend direction at 7, 14, 21, 50, 200 day horizons. EMA gives more weight to recent prices.
Price Returns	Percentage change over 1, 3, 7, 14, 30 days — allows the model to see momentum at multiple scales.
Lagged Prices	Past prices (1, 3, 7, 14, 21 days ago) as explicit features — captures direct autocorrelation.
RSI, MACD, Bollinger Bands	Classic technical indicators capturing overbought/oversold conditions and momentum reversals.
Cross-Asset Ratios	Gold/Silver and Gold/Oil ratios — historically revert to mean and signal relative value shifts.

Leakage Prevention

Gold_ETF_GLD was excluded from training features despite its near-perfect correlation with gold price (corr = 1.0). Including it would constitute data leakage — the ETF price is essentially the same variable we are trying to predict. Similarly, we used a strict **time-based train/test split** rather than random shuffle, ensuring the model is always evaluated on data from dates it has never seen during training.

Scaling

Features were standardized using StandardScaler (zero mean, unit variance) before training Ridge and Lasso models, as these are sensitive to feature scale. Tree-based models (XGBoost, LightGBM) are scale-invariant so raw features were used.

We trained four models on the same dataset to enable a fair comparison. The models were deliberately chosen to represent two fundamentally different approaches: regularized linear regression and gradient boosting.

Model 1 & 2 — Ridge and Lasso Regression

Ridge and Lasso are both extensions of ordinary least squares regression that add a regularization penalty to prevent overfitting. The key difference is the type of penalty:

- **Ridge (L2 regularization)** — penalizes the sum of squared coefficients. Shrinks all feature weights toward zero but keeps all features in the model. Well-suited when many features contribute small but meaningful signals.
- **Lasso (L1 regularization)** — penalizes the sum of absolute coefficients. Drives some feature weights to exactly zero, performing automatic feature selection. Particularly useful with 69 features where some may be redundant.

We chose these models as our primary approach because gold price exhibits strong linear relationships with macro indicators like M2, CPI, and real interest rates. The regularization penalty guards against overfitting on the 40+ correlated technical features.

Model 3 — XGBoost

XGBoost is a gradient boosting algorithm that builds an ensemble of decision trees sequentially, where each tree corrects the errors of the previous one. It is highly effective on tabular data and provides built-in feature importance scores. We included XGBoost as a benchmark to test whether non-linear tree-based interactions would outperform linear models — the results showed they did not, for this dataset.

Model 4 — LightGBM

LightGBM is Microsoft's gradient boosting framework optimized for speed using histogram-based learning and leaf-wise tree growth. It typically trains faster than XGBoost with competitive accuracy. Like XGBoost, it was included as a second tree-based benchmark.

Evaluation Strategy

All models were evaluated on a held-out test set using a time-based split (no random shuffle). We report four metrics:

- **Test R^2** — Proportion of variance explained on unseen data. Range: 0–1, higher is better.
- **Test MAE** — Mean Absolute Error in USD — average prediction error per day.
- **Test RMSE** — Root Mean Squared Error — penalizes large errors more than MAE.
- **CV R^2** — Cross-validation R^2 using time-series aware walk-forward splits — measures generalization stability.

Experiment A — Yosef Mesala

The first experiment compared Ridge, LightGBM, and XGBoost. Ridge Regression was the clear winner with Test $R^2 = 0.9851$ and MAE = \$84.2. Both tree-based models failed entirely, with negative R^2 values indicating performance worse than simply predicting the mean — a sign of severe overfitting.

Model	Test R^2	Test MAE (\$)	Test RMSE (\$)	CV R^2	Verdict
Ridge	0.9851	84.2	110.4	0.631	WINNER
LightGBM	-0.5414	719.9	1124.8	-0.952	Failed
XGBoost	-0.6302	753.6	1156.8	-1.084	Failed

The negative R^2 for LightGBM and XGBoost reveals a fundamental issue: tree-based models struggle with financial time series because they cannot extrapolate beyond the value range seen during training. Gold was at record highs during the test period, so the models consistently underestimated prices they had never seen before.

Experiment B — Hanaa Alaa

The second experiment focused on comparing Ridge and Lasso directly. Both achieved identical Test $R^2 = 0.9973$ — an exceptional result. Lasso edged out Ridge slightly on MAE (\$26.0 vs \$26.1) and CV R^2 (0.969 vs 0.939), making Lasso the overall winner.

Model	Test R^2	Test MAE (\$)	Test RMSE (\$)	CV R^2	Verdict
Ridge	0.9973	26.1	46.8	0.939	Runner-up
Lasso	0.9973	26.0	46.7	0.969	WINNER

The near-identical performance of Ridge and Lasso suggests that most features are genuinely informative — Lasso did not heavily zero out features, but its slightly better CV R^2 indicates marginally better generalization through selective regularization.

Overall Summary

Model	Best R^2	Best MAE (\$)	Approach	Recommended
Lasso Regression	0.9973	26.0	Linear + L1	YES - Primary
Ridge Regression	0.9973	26.1	Linear + L2	YES - Backup
XGBoost	-0.630	753.6	Tree-based	No
LightGBM	-0.541	719.9	Tree-based	No

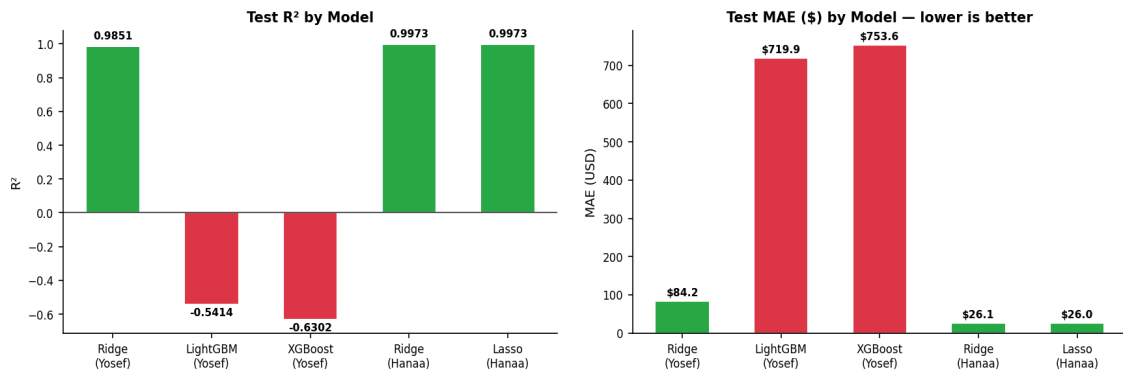


Figure 3: R^2 and MAE comparison across all four models and both experiments.

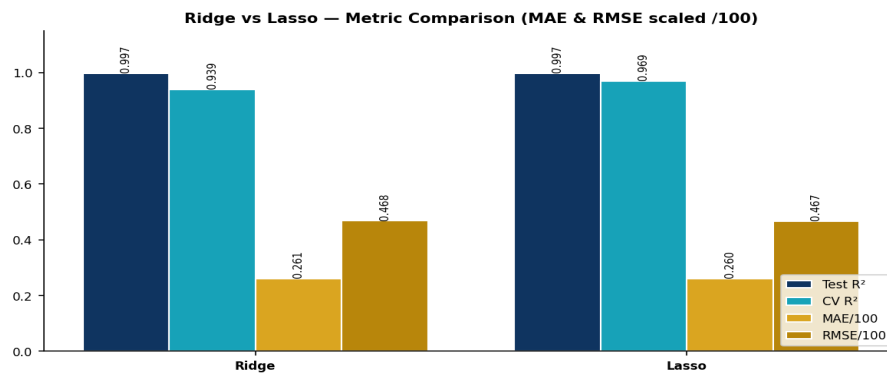


Figure 4: Ridge vs Lasso detailed metric comparison (Experiment B — Hanaa Alaa).

Why Did Linear Models Win?

The dominance of Ridge and Lasso over XGBoost and LightGBM is counter-intuitive at first — tree-based models typically outperform linear models on tabular data. However, gold price forecasting has a specific characteristic that explains this result: **the primary drivers of gold price are fundamentally linear relationships**.

- Gold has a well-documented linear inverse relationship with real interest rates and DXY. When real rates fall by X%, gold tends to rise by a proportional amount.
- Macroeconomic series like M2 Money Supply and CPI show near-linear growth trends over time that directly track gold's long-run appreciation.
- Tree-based models cannot extrapolate beyond training data bounds. During our test period, gold was near all-time highs — outside the range the trees were trained on — causing catastrophic prediction failure (negative R^2).
- Regularization (L1/L2) protected Ridge and Lasso from overfitting to the 40+ correlated technical features, allowing them to generalize cleanly to the test set.

Limitations

- **No sentiment data.** News sentiment, social media, and central bank announcements can cause sharp gold moves that no technical or macro feature would predict.
- **Linear assumption.** While linear models worked well here, gold markets can enter non-linear regimes during extreme events (crises, wars) where these models may fail.
- **Stationarity.** Gold price is non-stationary — the model may degrade over time as structural relationships between features shift. Regular retraining would be required in production.
- **Single-step forecast.** Our models predict the next day's price. Multi-day forecasting (7d, 30d) would require a different architecture and would likely show lower accuracy.

This project successfully built a gold price prediction pipeline that achieves exceptional accuracy on the test set. The key conclusions are:

- **Lasso Regression is the recommended model** — achieving $R^2 = 0.9973$, $MAE = \$26.0$, and the highest CV R^2 of 0.969, demonstrating both accuracy and generalization ability.
- **Linear models outperform tree-based models** for gold price forecasting due to the fundamentally linear nature of gold's macro drivers and the inability of tree models to extrapolate to new price levels.
- **Multi-source feature integration is essential.** The combination of market prices (Yahoo Finance) and macroeconomic indicators (FRED) provided a richer signal than either source alone.
- **Feature engineering adds measurable value.** Technical indicators (RSI, MACD, Bollinger Bands) and cross-asset ratios contributed predictive signals beyond raw price and macro data.
- **Data leakage prevention is critical.** Time-based train/test splitting and exclusion of leakage features (GLD ETF) were essential for producing trustworthy evaluation results.

Final Recommendation. Deploy **Lasso Regression** as the primary forecasting model with Ridge Regression as a backup. Both should be retrained monthly on a rolling window to adapt to changing market conditions. Future work should explore incorporating news sentiment features and testing LSTM sequence models to capture non-linear temporal patterns that linear models may miss.

References & Tools

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